Sweatshop on P-Chess: the v0.1.0 game-screen tickets

Resign button and Reason header (UI-013), draw offers (BOT-011), match score (UI-014), board resize grip (UI-015) and the repertoire snapshot file (HIST-003); GPT-6.1 Terra lineup, lane `v010`

Foreman, session `sweatshop/v010/2026-10-30-1831`

30 October 2026

Summary

A lane run (`-Lane v010`) in a fresh clone at `.artifacts-build/lanes/v010`, default lineup `terra`: stage 2 on `gpt-6.1-terra high`, hard tickets on `gpt-6.1-terra xhigh` (BOT-011), review on `gpt-6.1-sol xhigh`. The lanes `analysis` and `repertoire` ran at the same time in their own clones. Before launch the clone's gate passed (270 passed, 0 failed) with the Lc0 binary and Maia nets copied in. Launch at 18:31, last merge at 20:02.

- **Produced:** UI-013, BOT-011, UI-014, UI-015 and HIST-003 merged on the session, `Verdict: Approve`, all `Review: agent`. Session PR #27. Gate 287 passed, 0 failed (270 before the run).
- **Cost: $7.05** at API list price over the 8 priced stages (all reviews, `gpt-6.1-sol xhigh`); the 7 Terra stages log `?` (no Terra row in the driver's `$CodexPrices`). Codex quota: 6 points of the weekly window (Terra high 3, Terra xhigh 1, Sol 2, each ±1), in a window shared with the two concurrent lanes.
- **What went wrong:** UI-013 and UI-014 were reopened once each, two real P2 findings each. HIST-003's first review stopped on a scope question the driver does not relay to the proxy; the foreman asked it and relaunched.
- **Paused, by design:** the other nine `v010` tickets wait on the `analysis` lane: REV-005 on REV-002 and REV-003, REV-006, HIST-004 and UI-019 on REV-004, HIST-006 on REV-002, the rest behind HIST-004. They become runnable when PR #26 is merged and a new `v010` session starts.
- **Provisional verdict:** Terra is fast (median 8 min per first stage 2, 4–6 min per fix) and keeps test-first commits, but its first attempt on UI tickets misses state interactions around the feature. Under 10 reviews, an anecdote.

What was done

UI-013: a flag button between the clocks resigns the game; the first click arms it ("Click again to resign"), a second click within 2 s resigns, and it shares one arming window with `Esc`. Every saved PGN gains a `Reason` header (checkmate, resignation, time, ...). Main files `ui/main.gd`, `ui/main.tscn`, `core/pgn.gd`. Reopened once: the button showed `?` instead of a flag, because Terra wrote the glyph through a `python -c` edit inside PowerShell and the codepage replaced it (`ui/main.tscn`); and a clicked button kept keyboard focus, so with Settings open `Space` resigned the game behind the modal (`ui/main.gd`).

BOT-011: a ½ button offers Maia a draw. Maia answers with its next move from its own search evaluation (Lc0's `score cp`, verified first and recorded in the ticket) and a deterministic rule on material and its clock; "Draw agreed" ends the game as a draw by agreement (PGN `Reason: agreement`, stats count a draw), "Maia declined" plays its move. The button rests for ten of your moves after an offer. The rule's numbers live in the new `core/tunables.gd`. Main files `engine/uci.gd`, `bot/maia_player.gd`, `ui/main.gd`. Merged at the first review.

UI-014: rematches keep a match score: a win counts 1, a draw ½ each; the game-over panel shows "You 2 – 1 Maia 1500", and from the first rematch each clock row carries that player's score in a small

box that follows the colour swap. New game, a changed setting or an engine stop reset it. Main file `ui/main.gd`. Reopened once: after a finished rematch the boxes still showed the previous score (set only at game start), and they stayed visible in F11 focus mode, which shows only board and clocks (`ui/main.gd`).

UI-015: a grip on the board's bottom-right corner drags the board bigger or smaller; it stays square, is clamped to a minimum and to the window, the size is saved in the `board_size` setting, and a double-click returns to fit-to-window. Main files `ui/resize_grip.gd` (new), `core/settings.gd`. Merged at the first review.

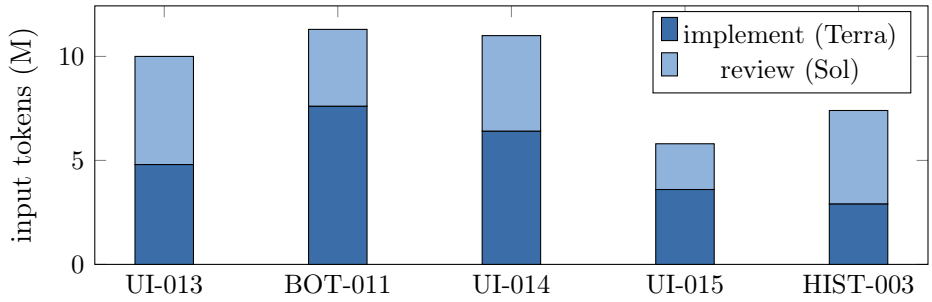
HIST-003: a game played with a repertoire loaded now saves the lines it was checked against beside its PGN, as `<game>.repertoire.json`, with the repertoire's colour, so a review opened later from the history (HIST-004) checks against that snapshot and not against today's repertoire. A failed snapshot write reports like a failed PGN write. Main file `core/history.gd`. No defect found; the first review stopped only to ask whether the reader function and the colour capture were inside the ticket's Primary files (they were in named files, required by criteria 4 and 1); the proxy approved and the re-review merged it.

Timeline

18:31	Clone of `origin` at `eea8c16`; gate 270/0; launch, pid 6924.
18:41	UI-013 to review (Terra high, 9 min).
18:46	UI-013 reopened, 2 findings. Stage 2 again.
18:50	UI-013 to review (4 min).
18:54	UI-013 merged. BOT-011 stage 2 (Terra xhigh, hard).
19:03	BOT-011 to review (9 min).
19:09	BOT-011 merged, 0 findings. UI-014 stage 2.
19:17	UI-014 to review (8 min).
19:22	UI-014 reopened, 2 findings. Stage 2 again.
19:28	UI-014 to review (6 min).
19:32	UI-014 merged. UI-015 stage 2.
19:40	UI-015 to review (8 min).
19:44	UI-015 merged, 0 findings. HIST-003 stage 2.
19:50	HIST-003 to review (6 min).
19:54	HIST-003 review ends at `blocked` (scope question, proxy unreachable from Codex). Driver: nothing runnable; PR #27 opened (4 tickets).
19:55	Foreman asks the proxy: Approve. Recorded on the session (`573eaa1`) and the ticket branch; `to-review` restored.
19:56	Relaunch, pid 30484.
20:02	HIST-003 merged, 0 findings; nothing runnable; PR #27 updated (5 tickets).

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
18:41	UI-013	impl	terra high	9	3.1M	97%	10k	?	to-review
18:46	UI-013	review	sol xhigh	5	3.4M	95%	14k	1.13	reopened
18:50	UI-013	impl	terra high	4	1.7M	94%	5k	?	to-review
18:54	UI-013	review	sol xhigh	4	1.8M	93%	10k	0.68	merged
19:03	BOT-011	impl	terra xhigh	9	7.6M	98%	19k	?	to-review
19:09	BOT-011	review	sol xhigh	6	3.7M	95%	15k	1.21	merged
19:17	UI-014	impl	terra high	8	3.7M	98%	10k	?	to-review
19:22	UI-014	review	sol xhigh	5	2.7M	94%	14k	0.96	reopened
19:28	UI-014	impl	terra high	6	2.7M	97%	6k	?	to-review
19:32	UI-014	review	sol xhigh	4	1.9M	91%	12k	0.79	merged
19:40	UI-015	impl	terra high	8	3.6M	98%	11k	?	to-review
19:44	UI-015	review	sol xhigh	4	2.2M	94%	13k	0.81	merged
19:50	HIST-003	impl	terra high	6	2.9M	96%	8k	?	to-review
19:54	HIST-003	review	sol xhigh	3	2.0M	92%	12k	0.80	blocked
20:02	HIST-003	review	sol xhigh	5	2.5M	94%	13k	0.66	merged
				86	45.5M		172k	7.05+?	



The chart is input tokens, not dollars: Terra's stages are unpriced. Wasted: the two reopen rounds (UI-013 3.1M + 3.4M, UI-014 3.7M + 2.7M in the rejected first attempt and its review) and HIST-003's blocked first review (2.0M in, $0.80), whose question needed no code change.

Comparison with previous runs

From `run-log.md`: Terra's first stage 2 on the four normal tickets took 6–9 min (median 8), the fixes after a reopen 4–6 min. Over its P-Chess history `gpt-6.1-sol high` took a median of 11 min per first stage 2 (21 runs, 0 failed). The hard ticket on `gpt-6.1-terra xhigh` took 9 min, against 11–17 min for the three hard tickets on `gpt-6.1-sol xhigh` in the 27 October QA lane. Token volumes are comparable (Terra 1.7–7.6M in, 5–19k out; Sol high 2–5M in, 10–28k out). Reviews cost about the same as on 27 October ($0.66–1.21 now, $0.75–0.79 then); the dearest are the reopening reviews and BOT-011's. The same-day `repertoire` lane (also Terra) reopened 1 of 3 tickets, with four findings.

Observations per model

- **gpt-6.1-terra high, stage 2** (6 stages): every ticket opens with a `test:` commit before its production commit, and both reopen fixes started with a red regression test that followed the reviewer's request (UI-013's dispatches real viewport input). Diffs stay small (+59 to +129 lines) and fixes are tightly scoped (+33 and +46 lines), with no scope creep. It never asked the proxy, never failed the gate, and disclosed manual acceptance honestly in every handoff. Weak spots: (1) *state around the feature*: UI-013 missed the Settings modal, UI-014 the result update and F11, all paths the repo already guards for other controls; (2) *non-ASCII through the shell*: UI-013's flag became `?`; it caught the same trap itself on BOT-011 (a label test, then a fix) and UI-014

(code points), but UI-014's fix commit and its later score-box commit carry no test of their own; (3) UI-015's grip-interaction test came after its feature commit (proved by a disclosed mutation check, not a red run); (4) HIST-003 widened the Primary-file scopes silently; a one-line note in the handoff would have spared the blocked review. Branch names vary per ticket (`ui-013`, `BOT-011-draw-offer`, `UI-014-match-score`, `HIST-003`); the helper accepts all of them.

- **gpt-6.1-terra xhigh, stage 2** (BOT-011, 1 stage): the strongest stage of the run: six red/green pairs, the ticket's "verify first" step on Lc0's `score cp` done and recorded before the rule, merged with 0 findings in 9 min.
- **gpt-6.1-sol xhigh, review** (8 stages, 3–6 min): each review checked the red/green order and reused the gate receipt after checking hash and code; both reopening reviews reproduced their findings with headless probes through real input, and both re-reviews raised nothing new (no drip-feed). The only code written in review was a comment (BOT-011's reason list), so no round-0 rows. HIST-003's first review stopped correctly on a contract question but could not reach the proxy.

Driver and runtime issues

1. `$CodexPrices` has no `gpt-6.1-terra` row: every Terra stage logs `?`, so the lineup's cost still cannot be compared (also reported by the `repertoire` lane).
2. A stage-3 question is not relayed to the proxy: the driver puts stage-2 questions to the proxy, but a review that ends `blocked` with the proxy unreachable (Codex sandbox) is left "for a human". Here the foreman asked and relaunched (one extra driver start, about 2 min).
3. Codex stages' `.txt.err` files fill with "Reading additional input from stdin..."; noise, no effect on the run.
4. Shutdown 1.0–2.6 s per stage; no crash, no timeout, no retry.

Recommendations

1. Add a `gpt-6.1-terra` row to `$CodexPrices` before the next Terra run.
2. Have the driver relay a blocked stage-3 question to the proxy as it does for stage 2 (twice per ticket per run), so a scope question does not end the run.
3. For UI tickets, add to the ticket's test list the state paths around the new control: Settings open, F11 focus mode, rematch and game end. Both reopens of this run, and OPEN-007's in the `repertoire` lane, were of that kind.
4. Keep Terra as default for `normal` tickets until ten reviews: scoreboard row `gpt-6.1-terra high` → `gpt-6.1-sol xhigh`, 9 reviews, 3 reopened (33%), S2 rate 33% — an anecdote, against `gpt-6.1-sol high` → `gpt-6.1-sol xhigh`, 18 reviews, 22%. Terra is faster per stage; whether it is cheaper waits on recommendation 1.
5. Before a release, play the manual checks below; every ticket of this run left product acceptance pending.

Human debt and proxy decisions

No `Débito humano:` lines in this run's tickets. One proxy decision, on HIST-003, asked by the foreman after the review blocked:

- `Proxy decided (opus-5.5 xhigh)`: "Approve. Criterion 4 requires the reader and criterion 1 requires the colour, so both edits are authorized. Stage 1 does not need to clarify anything first. Fold the widened scopes below into Primary files so the boundary matches what was merged - both edits sit in files the ticket already names, and the numbered criteria require them."

Reopen attribution

Labelled by the foreman (opus-5.5); the reviewer was `gpt-6.1-sol xhigh`, so no self-judging. Rows appended to `.scratch/reopen-attribution.md`.

Ticket (round)	Finding	Label	Why
UI-013 (1)	Resign button shows `?` instead of a flag	S2-explícito/ código	criterion 1; the glyph was lost in the shell and the scene never read back
UI-013 (1)	Focused button + Settings open + `Space` resigns behind the modal	S2-implícito	regression of stage 2's own focusable button against the existing Settings barrier
UI-014 (1)	Clock-row boxes keep the previous score after a finished rematch	S2-explícito/ código	criterion 3 and the ticket's test list; the test checked the panel only
UI-014 (1)	Score boxes visible in F11 focus mode	S2-implícito	regression against spec focus mode, a convention other controls follow

Totals: S2-explícito/código 2, S2-implícito 2, S1 0, S3-ruído 0. No review miss, no review-induced churn, no fix made in review. HIST-003's blocked review is not a reopen and has no row. No session-review, release-audit or post-release finding: the Copilot reviewer on PR #27 did not run (out of quota), which says nothing about defects.

Scoreboard

`scoreboard.ps1` over the ClassyCall, EasyMath, PhysiSyst, SIGAA-ME and SynchroNice main checkouts, plus this lane's clone in place of P-Chess's main checkout (same merged run log, and it holds this run's attribution rows before they reach `main`). Moved since the `repertoire` report of the same day: implementer `gpt-6.1-terra high` 4 → 10 stage runs (9 to review, 1 failed, median 8 min); new row `gpt-6.1-terra xhigh`, 1 run; pair `gpt-6.1-terra high` → `gpt-6.1-sol xhigh` 3 → 9 reviews, 1 → 3 reopened, rate unchanged at 33% (anecdote); findings row 3 rounds, 5 S2-explicit code, 3 S2-implicit; new interventions row `proxy-decision (opus-5.5 xhigh)` 1.

Implementer	Stage runs	To review	Failed	Asked	Median min to review	Cost
gpt-6.1-sol high	64	59	0	5	9m	$49.27 (64/64 priced)
opus-5.5 medium	3	2	1	0	6m	$4.79 (3/3 priced)
opus-5.5	7	4	0	3	14m	$15.12 (7/7 priced)
gpt-6-luna max	89	61	10	18	17m	$4.91 (85/89 priced)
sonnet-5.5 high	2	2	0	0	7m	$3.77 (2/2 priced)
gpt-6-astra high	1	0	0	1	-	$3.92 (1/1 priced)
opus-5.5 xhigh	18	14	4	0	12m	$91.78 (18/18 priced)
gpt-6.1-sol max	6	5	0	1	24m	$12.12 (6/6 priced)
haiku-5.5 high	2	0	2	0	-	$1.00 (2/2 priced)
opus-5.5 high	20	13	6	1	6m	$37.46 (20/20 priced)
gpt-6.1-sol xhigh	9	9	0	0	12m	$9.84 (9/9 priced)
gpt-6.1-terra high	10	9	1	0	8m	?
gpt-6.1-terra xhigh	1	1	0	0	9m	?
gpt-6-sol xhigh	1	1	0	0	9m	$0.40 (1/1 priced)
fable	1	1	0	0	11m	?
gpt-6-luna	8	4	4	0	37m	?
gpt-6-sol	6	5	1	0	17m	?
gpt-5.6-sol	5	4	1	0	17m	?
gpt-6-astra	18	11	6	1	8m	?
sonnet	5	3	2	0	14m	$10.53 (4/5 priced)
sonnet-5 xhigh	11	6	2	3	8m	$18.09 (11/11 priced)

Implementer	Reviewer	Reviews	Reopened	Reopen rate	Reopened for S2	S2 rate	Unclassified	Fixed in review for S2
gpt-6.1-sol high	gpt-6.1-sol max	38	12	32%	9	>=24%	2	1
opus-5.5 medium	gpt-6.1-sol max	2	1	50%	0	>=0%	1	0
opus-5.5	opus-5.5	4	0	0%	0	0%	0	0
gpt-6-luna max	opus-5.5 high	18	1	6%	1	6%	0	0
gpt-6-luna max	gpt-6-sol max	43	23	53%	22	51%	0	0
?	gpt-6-sol max	1	0	0%	0	0%	0	0
sonnet-5.5 high	gpt-6-sol max	2	1	50%	1	50%	0	0
opus-5.5 xhigh	gpt-6.1-sol max	6	3	50%	3	50%	0	0
?	gpt-6.1-sol max	1	0	0%	0	0%	0	0
gpt-6.1-sol max	gpt-6.1-sol max	4	1	25%	1	25%	0	0
?	opus-5.5 xhigh	4	1	25%	1	25%	0	0
opus-5.5 xhigh	opus-5.5 xhigh	7	0	0%	0	0%	0	0
opus-5.5 high	opus-5.5 xhigh	10	2	20%	2	20%	0	3
gpt-6.1-sol high	gpt-6.1-sol xhigh	18	4	22%	4	22%	0	0
gpt-6.1-sol xhigh	gpt-6.1-sol xhigh	9	1	11%	1	11%	0	1
gpt-6.1-terra high	gpt-6.1-sol xhigh	9	3	33%	3	33%	0	0
?	gpt-6.1-sol xhigh	1	0	0%	0	0%	0	0
gpt-6.1-terra xhigh	gpt-6.1-sol xhigh	1	0	0%	0	0%	0	0
gpt-6-sol xhigh	gpt-6-astra high	1	0	0%	0	0%	0	0
fable	fable	1	1	100%	0	>=0%	1	0

```
| ? | opus | 1 | 0 | 0% | 0 | 0% | 0 | 0 |
| gpt-6-luna | gpt-6-sol | 4 | 4 | 100% | 0 | >=0% | 4 | 0 |
| gpt-6-sol | gpt-5.6-sol | 5 | 4 | 80% | 0 | >=0% | 4 | 0 |
| gpt-5.6-sol | gpt-6-astra | 4 | 3 | 75% | 0 | >=0% | 3 | 0 |
| gpt-6-astra | gpt-5.6-sol | 11 | 5 | 45% | 0 | >=0% | 5 | 0 |
| sonnet | opus | 3 | 1 | 33% | 1 | 33% | 0 | 0 |
| sonnet-5 xhigh | opus-5.5 xhigh | 5 | 0 | 0% | 0 | 0% | 0 | 0 |

| Implementer -> Reviewer | Rounds | S2 explicit code | S2 explicit test | S2 implicit | S1 | S3 noise |
|---|---|---|---|---|---|---|
| gpt-6-astra medium -> gpt-6.1-sol max | 8 | 11 | 1 | 4 | 1 | 0 |
| gpt-6.1-sol-fast high -> gpt-6.1-sol-fast max | 2 | 2 | 0 | 0 | 0 | 0 |
| gpt-6.1-sol-fast max -> gpt-6.1-sol-fast max | 6 | 2 | 0 | 3 | 1 | 0 |
| gpt-6.1-sol high -> gpt-6.1-sol max | 26 | 16 | 4 | 23 | 1 | 0 |
| gpt-6-luna max -> opus-5.5 high | 1 | 1 | 1 | 0 | 1 | 0 |
| gpt-6-luna max -> gpt-6-sol max | 23 | 31 | 3 | 23 | 12 | 15 |
| sonnet-5.5 high -> gpt-6-sol max | 1 | 3 | 0 | 2 | 0 | 0 |
| opus-5.5 xhigh -> gpt-6.1-sol max | 3 | 3 | 0 | 2 | 1 | 0 |
| gpt-6.1-sol max -> gpt-6.1-sol max | 5 | 7 | 0 | 5 | 0 | 0 |
| haiku-5.5 high -> gpt-6.1-sol max | 3 | 3 | 1 | 5 | 1 | 0 |
| gpt-6.1-sol high -> gpt-6.1-sol xhigh | 53 | 53 | 4 | 23 | 3 | 0 |
| gpt-6.1-sol max -> gpt-6.1-sol xhigh | 2 | 0 | 0 | 3 | 0 | 0 |
| opus-5.5 high -> opus-5.5 xhigh | 8 | 7 | 0 | 3 | 0 | 0 |
| gpt-6.1-sol xhigh -> gpt-6.1-sol xhigh | 3 | 4 | 0 | 1 | 0 | 0 |
| opus-5.5 high -> gpt-6.1-sol xhigh | 1 | 2 | 0 | 0 | 0 | 0 |
| gpt-6.1-terra high -> gpt-6.1-sol xhigh | 3 | 5 | 0 | 3 | 0 | 0 |
| sonnet-5.5 high -> gpt-6.1-sol max | 3 | 5 | 0 | 2 | 3 | 0 |
| sonnet-5.5 xhigh -> gpt-6.1-sol max | 3 | 7 | 0 | 3 | 2 | 0 |
| sonnet-5.5 high -> opus-5.5 xhigh | 3 | 1 | 2 | 0 | 0 | 0 |
| gpt-6-astra medium -> gpt-6-astra xhigh | 1 | 1 | 0 | 0 | 0 | 0 |
| fable -> ? | 4 | 5 | 1 | 3 | 1 | 0 |
| sonnet -> ? | 10 | 3 | 6 | 9 | 3 | 2 |
| opus -> ? | 1 | 3 | 0 | 0 | 0 | 0 |
| sonnet? -> ? | 11 | 3 | 4 | 12 | 3 | 5 |
| sonnet -> opus | 1 | 1 | 1 | 0 | 0 | 0 |
| gpt-6-astra low -> gpt-6.1-sol max | 1 | 2 | 0 | 1 | 0 | 0 |

Evidence coverage: legacy flags and absent records are unknown, not zero defects or zero interventions.
| Discovery | Recorded findings | Earlier review miss (yes / known) | Review-induced churn (yes / known) |
|---|---|---|---|
| ticket-review | 372 | 5 / 177 | 1 / 179 |
| session-review | 28 | 22 / 24 | 0 / 27 |
| release-audit | 2 | 2 / 2 | 0 / 2 |
Runtime completion to process exit: 240.316s total over 196 measured stages; 333 unmeasured.
| Intervention | Recorded events |
|---|---|
| human-rescue | 30 |
| automatic-recovery | 89 |
| proxy-decision (unknown) | 37 |
| routine-approval | 18 |
| proxy-decision (gpt-6-astra high, fallback) | 58 |
| proxy-decision (opus) | 42 |
| proxy-decision (supervisor) | 12 |
| proxy-decision (opus-5.5 xhigh) | 1 |
```

Pending for the human

1. UI-013: resign with two clicks; let the hint expire and check one click only arms; click then `Esc` and `Esc` then click; open Settings after arming and press `Space` (nothing happens); check the flag renders.
2. BOT-011: offer a draw in the opening (declined) and in a dead-equal endgame with low clocks (accepted); check the $\frac{1}{2}$ button, the notice and the side panel.
3. UI-014: win, rematch, draw, rematch: the panel reads $1\frac{1}{2} - \frac{1}{2}$ and the boxes follow the colours; F11 hides them.
4. UI-015: drag the grip, restart, check the saved size; double-click returns to fit-to-window; check the cursor and the layout.
5. HIST-003: nothing to play now; its product check is HIST-004's criterion 6.
6. Review and merge session PR #27. Then merge `analysis`'s PR #26 and relaunch the `v010` lane for the nine waiting tickets, and tell the other open lanes' foremen to sync.